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The WEB-FOSS-NL project: open Source Tools for URL Finding and Information Extraction, Statistical Scraping Interest Group (SSIG)

Deliverable 4.1: part 1: Report on SSIG2

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Statistical Scraping Interest Group

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1. Introduction

The Statistical Scraping Interest Group (SSIG) was founded in 2025 to continue the successful network established during the Eurostat Web Intelligence Network (WIN) project where many statistical institutes followed the activities on analyzing, processing and interpreting web data for official statistics.

The group and the meetings are co-organized by Statistics Netherlands and Statistics Austria. The first took place 16-17 September 2025 in Vienna, Austria at Statistics Austria. The second took place 15-16 April 2026 in The Hague, the Netherlands at Statistics Netherlands. There are in total four SSIG meetings planned, the third meeting will be held in Porto 16-17 September 2026.

The SSIG functions as an informal community of practice: share experiences, learn from examples, explore new approaches. The group has the long-term goal to develop a formal statistical scraping theory.

The SSIG meetings provide the opportunity for presentations but also give plenty of time for discussion and general exchange.

All presentations of the meetings are publicly available on Github https://github.com/SNStatComp/SSIG .

2. Participants

FIRSTNAME	LASTNAME	Institution	Country
Olav	ten Bosch	Statistics Netherlands	NL
Alexander	Kowarik	Statistics Austria	AT
Christian	Salwiczek	Statistik Nord	DE
Luuk	Haarman	Statistics Netherlands	NL
Dominik	Blatt	Statistics Netherlands	NL
Sarah Valentina	Möller	Statistics Denmark	DK
Dennis	Pipenbring	Statistics Denmark	DK
Carlo	Bottai	University of Milano-Bicocca	IT
Tobias	Gramlich	Statistik Hessen	DE
Luca	Gramaglia	European Commission - Eurostat	EC
Femke	Bosman	Statistics Netherlands	NL
Johannes	Gussenbauer	Statistics Austria	AT
Leonardo	Castro Gonzalez	University of Bristol	GB
Michael	Reusens	National Bank of Belgium	BE
Johanna	Bolli-Kemper	University of Zurich	CH
Michael	Granitzer	University of Passau	AT
Arjen	De Vries	Radboud University, Nijmegen	NL
Djoerd	Hiemstra	Radboud University, Nijmegen	NL
Gijs	Hendriksen	Radboud University, Nijmegen	NL

Michael Granitzer presented online

3. Agenda

15 April at 9:30

9:00 - 9:30 Registration open
9:30 - 10:00 Start, welcome, SSIG in brief, who is here
Olav ten Bosch, Statistics Netherlands & Alex Kowarik, Statistics Austria

10:00 – 11:00 The Open Web Search project (remote)
Michael Granitzer, Univ. of Passau, AT online slides

11:00 – 11:30 Coffee Break

11:00 - 12:30 Presentations:

- Statistical scraping in Austria: WEB-FOSS-AT
  Alex Kowarik, Johannes Gussenbauer, Statistics Austria, AT
- Statistical Scraping in Netherlands: WEB-FOSS-NL
  Olav ten Bosch, Dominik Blatt, Luuk Haarman, Statistics Netherlands, NL

12:30-13:30 Lunch break

13:30- 14:30 Presentations:

- Integration of Company Websites into Business Registers: Identification, Quality Assurance, and Analytical Use of URLs
  Dennis Pipenbring and Sarah Moller, Statistics Denmark, DK
- Statistical Scraping at Statistics Hesse: past, present, future
  Tobias Gramlich, Statistics Hesse, DE

14:30-15:00 Coffee Break

15:00-16:30 Presentations:

- An Atlas of Economic Activities in the UK: tapping into web archives for social science research
  Leonardo Castro Gonzalez, Univ. of Bristol, UK
- Mapping the Italian Third Sector: Insights from Web-augmented Administrative Data
  Carlo Bottai, Univ. of Milano-Bicocca, IT
- How representative are queries of information retrieval test collections? (and how to use the web to collect queries and query frequencies for a certain market)
  Djoerd Hiemstra, Radboud University, NL

16:30-17:00 Discussion, closing first day (all)

18:30 Social Dinner (own expenses)

16 April at 9:30

9:00 - 9:30 Registration open
9:30 - 10:00 Presentation:

- A Personal Identifying Information (PII) redactor service in the Web Intelligence Hub (WIH)
  Luca Gramalia, Eurostat

10:00 - 11:00 The Open Web Index
Arjen de Vries, Radboud University, NL

11:00 – 11:15 Coffee Break

11:15 - 12:15 Presentations:

- From Job Postings to Vacancies: Addressing Non-Representativeness in Job Board Data
  Johanna Bolli-Kemper, Univ. of Zurich, CH
- From news feeds to signals: Ongoing research on Complementing business-cycle surveys with text indicators
  Michael Reusens, National Bank of Belgium, BE

12:15-12:45 Discussion, feedback, next meeting, closing

4. Meeting summary

1. Start, welcome, SSIG in brief, who is here (Olav ten Bosch)

Participants introduced their selves. Olav ten Bosch presented a very brief introduction in the concept of statistical scraping (bulk scraping versus statistical scraping), as well as the role of the Statistical Scraping Interest Group (SSIG).

2. The Open Web Search project (Michael Granitzer, University of Passau)

In this presentation Michael Granitzer introduced the objectives and future trajectory of the **OpenWebSearch.EU** project and its possible use for official statistics. The presentation highlights the critical need for a decentralized, publicly funded web index to counter the dominance of commercial "gatekeeper" search engines. By providing an open infrastructure, the project aims to foster digital sovereignty and transparency. The OWI serves as a "catalog" of the web that allows organizations—ranging from startups to government agencies—to build custom search engines and data products without relying on proprietary black-box algorithms.

The setup of the project and the partners involved are explained, i.e. CERN and three other computing centers. The Open Web Index would allow companies to provide services without the large upfront costs for setting up their own infrastructure for scraping and indexing. Specific open web index slices can be defined for targeted use. Moreover, all software is open source, allowing the use of open web search and indexing building blocks at other institutes.

As the OpenWebSearch.EU project officially concluded in February 2026, the slides outline the transition to the **European Open Web Search Initiative**. The goal is to sustain the OWI as a permanent public infrastructure. The current facilities are about 20 times less in power and size then commercial alternatives. The presentation emphasizes that an open web index is not just a technological tool, but a foundational requirement for a fair digital economy and an informed society, enabling "search science" to advance independently of commercial interests. The discussion following the presentation focuses on possible use for official statistics. Would a high-quality slice of OWS web data be achievable and useful?

3. Statistical scraping in Austria: WEB-FOSS-AT (*Alex Kowarik, Johannes Gussenbauer, Statistics Austria*)

The Austrian presentation introduced the current progress of the WEB-FOSS-AT project, focusing on the development of the open-source R package taRantula and practical use cases for statistical web scraping. The R package taRantula, which evolved from a collection of scripts into a standalone, modular scraping framework. The package supports both URL discovery (currently via Google Custom Search) and scalable scraping using Docker-based environments with Selenium or lightweight HTTP requests. Results are stored in DuckDB, enabling efficient querying and reproducibility through snapshotting.

In addition to the technical framework, two applied use cases were presented. One example focuses on scraping wine prices from Austrian producer websites to derive input for agricultural price statistics. This involves selecting enterprises from the business register, finding the URL to their websites, and extracting structured information such as prices, volumes, and product characteristics, partly supported by large language models. Another use case addresses the classification of economic activities (NACE), where web content is scraped and analysed to support reclassification efforts under the updated NACE classification.

The presentation highlighted both the potential and current limitations of scraping, including challenges in URL identification, scraping stability, and model accuracy. Future development priorities include improving robustness, extending functionality, exploring alternative data sources such as the Open Web Index, and preparing the package for broader dissemination (CRAN release towards the end of the project).

4. Statistical Scraping in Netherlands: WEB-FOSS-NL (Olav ten Bosch, Dominik Blatt, Luuk Haarman, Statistics Netherlands)

The project WEB-FOSS-NL aims to deliver open-source building blocks for statistical scraping. It explores statistical scraping in general sense, with a particular application focus on job vacancy data for this project. It combines business register information with methods like URL finding, targeted web scraping, and text analysis with LLMs to extract relevant information on job vacancies (or other variables) from company websites. The workflow involves identifying enterprise websites, selectively scraping pages likely to contain job advertisements, and using machine learning models to classify relevant content.

In the first year the general URL finding package, https://github.com/snstatcomp/urlfinding , was improved. The number of queries necessary for the Google search were reduced and optimised. Two new releases were produced.

Moreover, in the first year a test run was performed, on a test set from the business register, with URL finding and focussed scraping and model-based text analysis. Data from Eurostat Web Intelligence Hub was used to train a model with positive cases (OJAs). Combined with negative cases (Wikipedia data) a first model was created. Tests showed the need for further model refinement, which was performed by continuous retraining. Minimal finetuning was enough to improve model accuracy significantly. The results are promising. Future improvements include better scraping tools, more robust classifiers, and scaling up the approach.

5. Integration of Company Websites into Business Registers: Identification, Quality Assurance, and Analytical Use of URLs (Dennis Pipenbring and Sarah Moller, Statistics Denmark)

This presentation from Statistics Denmark demonstrated how National Statistical Institutes (NSIs) can systematically implement, maintain, and quality-assure company URLs within their National Statistical Business Registers (NSBRs). The presentation started by expressing the dream of a 'statistical cookie' on every website with the information for official statistics. From there, it provided the methodology developed by Statistics Denmark. First, it described how company URLs are identified through a multi-source approach that combines existing register information in the ABR with data from the national top-level domain register to ensure broad and consistent coverage. These data are supplemented by the use of a locally hosted Llama large language model (LLM), which analyses company names, legal identifiers, and auxiliary register attributes to identify and suggest potential URLs. Second, it outlined the quality assurance framework applied to the identified URLs, including automated checks, and procedures for accurately linking URLs to the correct enterprises and legal units. Third, it presented the preliminary development of content-based indicators derived from company websites, which are used to segment URLs according to observable characteristics and business activities. In this context, a locally hosted Llama

large language model (LLM) is also applied to analyse website content and support the segmentation process. Statistics Denmark participates in the Eurostat-funded OBEC project, where these web-based data sources are used to train a machine-learning model for the classification of companies' economic activities based on NACE Rev. 2.1. In addition, the approach supports analytical segmentation of enterprises and contributes to reducing respondent burden by exploiting information already available online.

6. Statistical Scraping at Statistics Hesse: past, present, future *(Tobias Gramlich, Statistics Hesse, DE)*

Statistics Hesse began its webscraping journey in 2018 by leveraging the Google Custom Search API and participating in collaborative ESSnet initiatives like the Web Intelligence Network. As of June 2025, the office maintains a full production system that identifies URLs for Statistical Business Register (SBR) units, monitors keyword occurrences, and automatically extracts managing director information from trade registry documents. This robust infrastructure, running on a farm of six servers, is built on a technical stack featuring Shinyproxy for the user interface, Keycloak for authentication, and a combination of R and Python to manage webscraping processes and database synchronization.

Looking ahead, the office plans to enhance its database by allowing users to upload verified URLs and implementing automated re-checks every twelve months to maintain data quality. To capture data on businesses missing from official registries, future efforts will focus on extracting legal imprint information directly from company websites, as mandated by the Telemedia Act. A critical strategic shift involves transitioning from Google's API to the OWS/OWI framework to bypass the current 10,000-search daily limit and reduce reliance on non-EU commercial providers. This evolution aims to streamline operations while ensuring greater technical independence and results comparability.

7. An Atlas of Economic Activities in the UK: tapping into web archives for social science research (*Leonardo Castro Gonzalez, Univ. of Bristol, UK*)

The challenge to analyse and model economic activities across time and space is key to answering economic geography problems. However, building and accessing up-to-date and dynamic tools to study how a country's economy evolves is a major challenge. In the UK, e.g., the Standard Industrial Classification (SIC) has not been updated since 2007, making it impossible to study the evolution of economic activities. This presentation shows an attempt to create an open-source dynamic Atlas of economic activity for the UK at postcode-level granularity, to answer questions such as: what are spatial footprints and dynamics of economic activities in the UK? Data sources used include Common Crawl in combination with LLMs. The results was validated against the proprietary FAME database. The presentation reported on the first findings and plans for disseminating the results.

8. Mapping the Italian Third Sector: Insights from Web-augmented Administrative Data (*Carlo Bottai, Univ. of Milano-Bicocca, IT)*

In this presentation Carlo Bottai presented the following case: reliable data on the Italian Third Sector remain limited despite its socio-economic importance. Therefore this paper addresses a methodology for assembling a comprehensive dataset on the sector by systematically exploiting administrative data from

the Single National Register of the Third Sector (RUNTS). The proposed pipeline automates the acquisition and cleaning of data on more than 134,000 organisations through a tailored web-scraping procedure. To address common shortcomings of administrative data, such as incompleteness and self-reporting bias, a novel web-augmentation strategy is implemented, combining information extracted from organisational websites with a Generative AI–based classification system. This approach allows the identification of beneficiaries, territorial scope, and economic activities, and enables comparison between web-derived and self-declared information for the latter case. Using data for the second quarter of 2025, the paper provides an up-to-date picture of the Italian Third Sector and offers a new informational base for socio-economic research and co-programming of local policies.

9. How representative are queries of information retrieval test collections? (and how to use the web to collect queries and query frequencies for a certain market) (*Djoerd Hiemstra, Radboud University, NL)*

The presentation examines whether queries used in information retrieval test collections are representative of real user queries. Test collections typically include a small, curated set of queries, while real-world query logs (such as AOL data) show highly skewed distributions, where a few queries dominate frequency (power law distribution). Using statistical methods like stratified sampling and chi-square tests, the study finds that common test query sets are unlikely to be random samples of actual queries. This mismatch suggests evaluation results may not reflect real-world search behavior. The work proposes alternative approaches, such as sampling queries based on real frequencies and using web data and scraping techniques to build more realistic test collections.

10. A Personal Identifying Information (PII) redactor service in the Web Intelligence Hub (WIH) (*Luca Gramalia, Eurostat)*

Eurostat presented a specialized tool for identifying and removing Personally Identifying Information (PII) from web-scraped Online Job Advertisements. This project addresses the legal requirements of the GDPR, specifically the principle of data minimization and the appropriate safeguards needed for statistical processing under Article 89. The redactor is designed to identify and remove sensitive data points such as individual names, social media profiles, phone numbers, and email addresses that are often captured during the scraping of public job portals. By redacting this information immediately after collection, Eurostat ensures the lawful and ethical use of big data for tracking labor market demands.

The technical architecture relies on the open-source Microsoft Presidio framework, which integrates multiple detection methods including regular expressions and machine learning. For structured data like URLs and emails, the system uses regex-based recognizers, while for names, it leverages the multilingual XLM-RoBERTa model capable of processing over 100 languages. The tool is packaged as a Docker container and operates as a REST API, enabling it to process various input formats including plain text, JSON, and HTML. Future efforts focus on validating redaction accuracy using synthetic samples and evaluating the redactor's impact on downstream data classification tasks before full-scale production deployment.

11. The Open Web Index (Arjen de Vries, Radboud University, NL)

Arjen de Vries presented the Open Web Index (OWI) specifics of the OpenWebSearch.eu initiative (see presentation Michael Granitzer above). By creating an Open Web Index (OWI) based on European values and jurisdiction, the project provides researchers with unbiased, transparent access to web data. Arjen de Vries provided a general background in web indexing and explained on various more technical aspects, such as the core concept of an inverted index, a highly efficient & flexible data structure for fast retrieval, a technique to rank documents efficiently on similarity to a query.

Various search engines have been implemented on top of the index as an example, but a more natural use is to work on the index itself. CIFF formats are available for download and can be merged with content from institutes to provide for their own search services. The presentation highlighted the integration of the Open Web Index with DuckDB to facilitate efficient remote querying. By utilizing DuckDB's analytical capabilities, users can process vast amounts of web data locally or via federated networks, effectively pushing search functionality to the edge. This approach decouples the search process from centralized indexes and allows for a privacy-preserving data analysis using local resources. The statistical community could be a good use case for use of the OWI.

12. From Job Postings to Vacancies: Addressing Non-Representativeness in Job Board Data (*Johanna Bolli-Kemper, Univ. of Zurich, CH)*

Johanna Bolli-Kemper from the University of Zurich addresses the non-representativeness of job board data when tracking labor market vacancies. As a non-probability sample, job board data contains systematic selection biases across sectors and firm sizes, failing to accurately reflect total economic demand. To mitigate this, the research evaluates weighting techniques like post-stratification and Inverse Probability Weighting (IPW).

The study finds that while post-stratification aligns data with known population margins, it misses firms without an online presence. IPW offers a more robust solution by estimating the propensity of firms to post online, though missing characteristics currently limit its full application. By utilizing "Overlap IPW"—comparing job board ads with firm website scrapes—the researchers approximate corrections for the online market. Ultimately, achieving full population representativeness requires integrating firm surveys and official statistical registers to bridge the gap between digital postings and total vacancies.

13. From news feeds to signals: Ongoing research on Complementing business-cycle surveys with text indicators (*Michael Reusens, National Bank of Belgium, BE)*

Michael Reusens from the National Bank of Belgium presents research on using news-based indicators to complement traditional business-cycle surveys. While surveys are trusted instruments for measuring economic expectations, they are limited by high costs, declining response rates, and a lack of qualitative narrative detail. To bridge these gaps, the bank utilizes high-frequency news data to capture economic sentiment in near real-time. These text-based signals provide a timely proxy for market trends and can potentially identify economic turning points earlier than conventional surveys.

A prototype index shows significant co-movement with consumer confidence since 1999, validating news as a powerful tool for economic monitoring. Despite this success, news indicators primarily track media narratives rather than direct household sentiment and lack the detailed disaggregation found in official surveys. Future efforts will focus on enhancing these models with large language models and exploring

additional web-based sources like job postings and online retail prices. This approach aims to create a more flexible and robust framework for real-time business-cycle assessment.

14. Discussions

This SSIG meeting contained quite some presentations from Statistical Institutes, universities and the WOS / OWI project, gaining general knowledge on where we could work together. The following practical points have been defined:

- One of the most promising starting points for the use of OWS / OWI for official statistics seems to experiment with ***URL finding***. Generally speaking, URL finding is linking web data, such as websites to statistical units, such as enterprises in a business register. The use of an open index for discovery, solves the possible problem of search engine leakage. Public index files (CIFF) can freely be merged with private information available at the statistical institute. Moreover, at the side of the OWS/OWI project index slices could be tailored towards specific use for statistical institutes in different countries. Also, the open-source software developed by the OWS/OWI project could be freely used, and possible improved, by the statistical institutes. On the side of the OWS / OWI follow-up the use of the index by official statistical institutes would imply a practical use case of serious governmental partners. Discussions an possible follow-up budget for such activities were initiated. In between SSIG2 and SSIG3 the SSIG management (Statistics Austria and Netherlands) will organise one online meeting to explore such possibilities in more detail with interested SSIG participants.

- In addition to the previous point, Arjen de Vries from OWS created a channel for this group, more details here:
  https://openwebsearch.eu/community/ows-eu-community-on-mattermost/
  https://openmm.it4i.eu/ows-community/channels/ssig-ows-project

- In limited time, some other discussions were touched upon, such as the idea to develop alternative, more web-oriented classifications.

- The idea for a joined /back-to-back meeting with all Eurostat web data grants recipients was discussed again. SSIG3 could be an option for such event, if it is extended to some extent. Statistics Austria and Eurostat will be in contact about this.